

Grove - Button
 Grove - LED
 Grove - Rotary Angle
 Grove - Sound Sensor
 Grove - Smart Relay
 Grove - Temperature
 Grove - Touch Sensor
 Grove - Light Sensor
 Grove - Mini Servo
 Grove - LCD RGB Backlight
 8GB Micro SD Card with an SD Card Adapter
 9V to Barrel Jack Adapter - 126mm
 26AWG Grove Cable
 Micro USB Cable
 Serial Cable
 Ethernet Cable

INTEL GALILEO

The Intel® Galileo Gen 2 board falls in the family of Arduino-certified development and prototyping boards based on Intel® architecture and specifically designed for makers, students, educators, and DIY electronics enthusiasts.



The Intel Galileo board

Providing users with a fully open source hardware and software development environment, the Intel Galileo Gen 2 board complements and extends the Arduino line of products to deliver more advanced compute functionality to those already familiar with Arduino prototyping tools. The Intel Galileo Gen 2 development board is designed to be hardware-, software-, and pin-compatible with a wide range of Arduino Uno® R3 shields and additionally allows users to incorporate Linux® firmware calls in their Arduino sketch programming.

FEATURE HIGHLIGHTS

- Intel® Quark SoC X1000 application processor, a 32-bit, single-core, single-thread, Intel® Pentium® processor instruction set architecture (ISA)-compatible, operating at speeds up to 400 MHz.
- Support for a wide range of industry standard I/O interfaces, including a full-sized mini-PCI Express® slot, 100 Mb Ethernet port,

microSD® slot, USB host port, and USB client port.

- 256 MB DDR3, 512 kb embedded SRAM, 8 MB NOR Flash, and 8 kb EEPROM standard on the board, plus support for microSD card up to 32 GB.
- Hardware and pin compatibility with a wide range of Arduino Uno R3 shields.
- Programmable through Intel's Arduino integrated development environment (IDE) that is supported on Microsoft Windows, Mac OS, and Linux host operating systems.
- Support for Yocto 1.4 Poky Linux release.

WHAT'S NEW WITH THE INTEL® GALILEO GEN 2 BOARD

- 6-pin 3.3V USB TTL UART header replaces 3.5 mm jack RS-232 console port for Linux debug. New 6-pin connector mates with standard FTDI® USB serial cable (TTL-232R-3V3) and popular USB-to-Serial breakout boards. 12 GPIOs now fully native for greater speed and improved drive strength.
- 12-bit pulse-width modulation (PWM) for more precise control of servos and smoother response.
- Console UART1 can be redirected to Arduino headers in sketches, eliminating the need for soft-serial in many cases.
- 12V power-over-Ethernet (PoE) capable (PoE module installation required).
- Power regulation system changed to accept power supplies from 7V to 15V.
- The Intel® IoT Developer Kit for Intel Galileo Gen 2 adds C, C++, Python, and Node.js/ Javascript support for developing connected sensor Internet-of-Things applications.
- Intel Galileo also supports the Wylodrin platform that provides Python and Node.js Programming environments from a remotely-connected browser.
- In addition to open source Yocto Linux, Intel Galileo Gen 2 supports VxWorks (RTOS), and now Microsoft Windows is supported directly from Microsoft.



The Intel Edison board

INTEL EDISON

Intel® Edison is a solution designed to lower the barriers to entry for quick prototyping and productizing the connected computing devices driving the next industrial revolution. This wave of connected devices will be a diversified collection of products invented for specific consumers and consumer needs.

The Intel Edison solution was designed specifically for the creators of these devices-emerging entrepreneurs eager to invent the future. It combines a small, powerful, adaptable hardware platform and partner-enabled ecosystem with extended software compatibility and supportive online environment. You can even take this to the next level with the help of open source expansion boards for Edison and utilize the community's inputs to bring the best of everything together.

FEATURE HIGHLIGHTS

- At the center of Intel Edison is an Intel® Atom™ 22nm SoC with a dual-core CPU at 500MHz and the Intel® Quark™ microcontroller at 100 MHz. This allows the product to collect and preprocess data via the MCU in a low power state and hand the filtered data off to the CPU for analytics. Edison includes 1GB of memory, 4GB of storage and dual-band Wi-Fi and Bluetooth 4.0 for communications. It supports 40 GPIOs with multiple configuration options
- High performance, dual-core CPU and single core micro-controller supports complex data collection in a low power package
 - Integrated Wi-Fi, Bluetooth Low-Energy (LE), memory, and storage simplifies configuration and increases scalability
 - 40 multiplexed GPIO interfaces with expansion board options for total project design and flexibility
 - The Intel® IoT Developer Kit for Intel Edison adds C, C++, Python, and JavaScript support for developing connected sensor Internet-of-Things applications.
 - Supports a variety of Integrated Development Environments such as Arduino®, Eclipse®, Wylodrin®, and Intel's XDK.

SOFTWARE IMAGE:

Intel also provides integration and availability of a wide variety of software to go with the hardware kits to enable the best out of IoT devices. Here is an overview of what you have access to: -

For Intel Galileo: Yocto 1.6 based embedded Linux boot image that includes helpful Linux tools